LESSON 11
GENERAL MULTIPLICATION

SUMMARY

11.1 Revision
11.2 Two-Figure Numbers – multiplying 2-figure numbers in one line, from left to right.
11.3 Moving Multiplier – multiplying long numbers by a 2-figure number.
11.4 Extension – multiplying 3-figure numbers.
11.5 Multiplying Binomials – using the same pattern.
11.6 Multiplying 3-Figure Numbers – extension of previous pattern.
11.7 Written Calculations – from left to right.

11.1 REVISION

We have seen various methods of multiplication but they were all for special cases, where some special condition was satisfied, like both numbers being close to 100 for example.
We come now to the general multiplication technique, by which any two numbers can be multiplied together in one line, by mere mental arithmetic.

First let us briefly revise how we multiply by a single figure number (as in Section 4.2).

You may wish to begin this lesson with written calculations rather than mental: if so go to Section 11.7, but you will need the methods described in Sections 11.2, 11.3, 11.6.

1 Find **74 × 8**.

We multiply each of the figures in 74 by 8 starting at the left:

7 × 8 = **56** and 4 × 8 = **32**.

These are combined by carrying the 3 in 32 over to the 6 in 56: 56,32 = 592.

The inner figures are merged together. So **74 × 8 = 592**.

2 Find **827 × 3**.

The three products are **24**, **6**, **21**.

The first two products are combined: 24,6 = 246 no carry here as 6 is a single figure, then 246 is combined with the 21: 246,21 = 2481. So **827 × 3 = 2481**.

3 Find **77 × 4**.

The products are **28**, **28**.

And 28,28 = 308 (the 28 is increased by 2 to 30). So **77 × 4 = 308**.

✐ **Practice A** Multiply the following mentally:

a 73 × 3 **b** 63 × 7 **c** 424 × 4 **d** 777 × 3

e 654 × 3 **f** 717 × 8 **g** 876 × 7

---

a 219 **b 441** **c 1 696** **d 2 331**
e 1 962 **f 5 736** **g 6 132**

11.2 TWO-FIGURE NUMBERS

The *Vertically and Crosswise* formula gives us the pattern for multiplying any numbers. For 2-figure numbers it works like this.

4 Find **21 × 23**.

Think of the numbers set out one below the other:

```
2  1
2  3  ×
4 8 3
```

There are 3 steps

A. Multiply vertically in the left-hand column: 2 × 2 = **4**, so 4 is the first figure of the answer.

```
2  1
|
2  3  ×
4
```

B. Multiply crosswise and add: 2 × 3 = 6, 1 × 2 = 2, 6 + 2 = **8**, so 8 is the middle figure of the answer.

```
2  1
  ×
2  3  ×
4 8
```

C. Multiply vertically in the right-hand column: 1 × 3 = **3**, 3 is the last figure of the answer.

```
2  1
   |
2  3  ×
4 8 3
```

5 Find **14 × 21**.

$$\begin{array}{r} 1\ \ 4 \\ \underline{2\ \ 1} \times \\ \underline{2\,9\ 4} \end{array}$$

A: vertically on the left: 1 × 2 = **2**,
B: crosswise: 1 × 1 = 1, 4 × 2 = 8 and 1 + 8 = **9**,
C: vertically on the right: 4 × 1 = **4**.

This is of course very easy and straightforward and is just mental arithmetic. We should now practice this *vertical and crosswise* pattern to establish the method.

✐ **Practice B** Multiply mentally:

a 2 2	**b** 2 1	**c** 2 1	**d** 2 2	**e** 6 1	**f** 3 2	**g** 3 1	**h** 1 3
3 1×	3 1 ×	2 2 ×	1 3 ×	3 1 ×	2 1 ×	3 1 ×	1 3 ×
____	____	____	____	____	____	____	____

a 682	**b 651**	**c 462**	**d 286**
e 1 891	**f 672**	**g 961**	**h 169**

CARRIES

The previous examples involved no carry figures, so let us consider this next.

6 Find **23 × 41**.

$$\begin{array}{r} 2\ \ 3 \\ \underline{4\ \ 1} \times \\ \underline{9\,4\,3} \end{array}$$

The 3 steps give us: 2 × 4 = **8**,
2 × 1 + 3 × 4 = **14**,
3 × 1 = **3**.

The 14 here involves a carry figure, so in building up the answer mentally from the left we merge these numbers as before.

The mental steps are: 8
8,14 = 94 (the 1 is carried over to the left)
94,3 = 943

So **23 × 41 = 943**.

7 Find **23 × 34**.

```
  2  3
  3  4 ×
  7 8 2
```

The steps are: 6
6,17 = 77
77,12 = **782**

8 Find **33 × 44**.

```
  3  3
  4  4 ×
 1 4 5 2
```

The steps are: 12
12,24 = 144
144,12 = **1452**

You can now multiply any two 2-figure numbers together in one line.

✐ **Practice C** Multiply the following mentally:

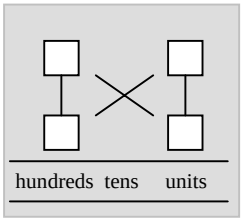
a 21 × 47	**b** 23 × 43	**c** 24 × 29	**d** 22 × 28	**e** 22 × 53	**f** 31 × 36
g 22 × 56	**h** 31 × 72	**i** 44 × 53	**j** 33 × 84	**k** 33 × 69	**l** 34 × 42
m 33 × 34	**n** 22 × 52	**o** 34 × 66	**p** 51 × 54	**q** 35 × 67	**r** 55 × 59
s 54 × 64	**t** 55 × 63	**u** 44 × 81	**v** 45 × 81	**w** 48 × 72	**x** 34 × 19

a 987	**b 989**	**c 696**	**d 616**	**e 1 166**	**f 1 116**
g 1 232	**h 2 232**	**i 2 332**	**j 2 772**	**k 2 277**	**l 1 428**
m 1 122	**n 1 144**	**o 2 244**	**p 2 754**	**q 2 345**	**r 3 245**
s 3 456	**t 3 465**	**u 3 564**	**v 3 645**	**w 3 456**	**x 646**

You may have found in this exercise that you prefer to start with the crosswise multiplications, and put the left and right vertical multiplications on afterwards.

EXPLANATION

It is easy to understand how this method works.
The vertical product on the right multiplies units by units and so gives the number of units in the answer. The crosswise operation multiplies tens by units and units by tens and so gives the number of tens in the answer. And the vertical product on the left multiplies tens by tens and gives the number of hundreds in the answer.



So this easy multiplication method, which is quite general, is also easy to understand. It can be done from left to right or right to left (see Section 11.7) it applies to algebraic expressions just as well (see Section 11.5) and it can be reversed to give a simple division method (see Section 16.4).

EXPLANATION OF EARLIER SPECIAL METHOD

We can now explain the special method of multiplication under *By One More than the One Before* from Section 10.2 for multiplying numbers like 72 × 78 in which the first figures are the same and the last figures add up to 10.

Using the present sutra for 72 × 78:

$$\begin{array}{r} 7\ 2 \\ \underline{7\ 8} \\ \underline{5\ 6\,{}_{7}1_{1}\,6} \end{array}$$

We see that the cross-product is eight 7's and two 7's, that is ten 7's, or 70. The zero here ensures that the 2-digit product 2 × 8 = 16 can go straight into the last two places, and this will always happen when the conditions for this type of product are met. The 7 in 70 means an extra 7 in the left-hand product: so there are eight 7's altogether.

As the method of squaring numbers that end in 5 is a special case of the above (see Section 12.1), this can also be explained this way.

11.3 MOVING MULTIPLIER

In multiplying a long number by a single figure, for example 4321 × 2, we multiply each of the figures in the long number by the single figure. We may think of the 2 moving along the row, multiplying each figure vertically by 2 as it goes.

9 Find **4321 × 32**.

4 3 2 1 **3 2**	Similarly here we put 32 first of all at the extreme left. Then vertically on the left, 4 × 3 = **12**. And crosswise, 4×2 +3×3 = **17**.
4 3 2 1 **3 2**	Then move the 32 along and multiply crosswise: 3×2 + 2×3 = **12**.
4 3 2 1 **3 2**	Moving the 32 once again: multiply crosswise, 2×2 + 1×3 = **7**. Finally the vertical product on the right is 1×2 = **2**.

These 5 results (in bold), 12,17,12,7,2 are combined mentally, as they are obtained, in the usual way:

12,17 = 137

137,12 = 1382

1382,7,2 = **138272**

So we multiply crosswise in every position, but we multiply vertically also at the very beginning and at the very end.

10 Find **31013 × 21**.

Here the 21 takes the positions:

3 1 0 1 3	**3 1 0 1 3**	**3 1 0 1 3**	**3 1 0 1 3**
2 1	**2 1**	**2 1**	**2 1**

The six mental steps give: 6,5,1,2,7,3
so the answer is **651273**.

Practice D Multiply using the moving multiplier method:

a 3 2 1
2 1

b 3 2 1
2 3

c 4 2 1
2 2

d 3 2 1
4 1

e 1 2 1 2
2 1

f 1 3 3 1
2 2

g 1 3 1 3
3 1

h 1 1 2 2 1
2 2

a	**6 741**	**b**	**7 383**	**c**	**9 262**	**d**	**13 161**
e	**25 452**	**f**	**29 282**	**g**	**40 703**	**h**	**246 862**

11.4 EXTENSION

11 Find **123 × 132**.

```
    1 2 3
    1 3 2 ×
  1 6 2 3 6
```

The *Vertically and Crosswise* formula can be extended to deal with this, but in fact the previous vertical/crosswise/vertical pattern can be used on this sum also.

We can split the numbers up into 12/3 and 13/2, treating the 12 and 13 as if they were single figures:

```
  12    3
  13    2
 162 3 6
```

Vertically 12 × 13 = **156**,
crosswise 12 × 2 + 3 × 13 = **63**,
vertically 3 × 2 = **6**.

Combining these mentally we get: 156
156,63 = 1623
1623,6 = **16236**.

Practice E Multiply, treating the numbers as 2-figure numbers:

a
```
 1 1 2
 2 0 3
 -----
```

b
```
 1 2 3
 1 3 1
 -----
```

c
```
 1 2 3
 1 2 2
 -----
```

d
```
 1 1 2
 1 2 3
 -----
```

e
```
 4 2 1
   2 2
 -----
```

a 22 736 **b 16 113** **c 15 006**
d 13 776 **e 9 262**

12 **304 × 412 = 125248**.

Here we may decide to split the numbers after the first figure: 3/04 × 4/12.

```
  3     04
  4     12
 12  52  48
```

When we split the numbers in this way the answer appears **two digits at a time**.

The 3 steps of the pattern are: 3×4 = **12**,
3×12 + 4×4 = **52**,
4×12 = **48**.

These give the 3 pairs of figures in the answer.

✐ **Practice F** Multiply using pairs of digits:

a 2 1 1 3 0 4	**b** 3 0 7 4 0 7	**c** 2 0 3 4 3 2	**d** 2 1 1 3 1 1
e 5 0 4 5 0 4	**f** 5 0 1 5 0 1	**g** 7 1 2 1 1 2	**h** 7 0 3 2 1 1

a 64 144	**b 124 949**	**c 87 696**	**d 65 621**
e 254 016	**f 251 001**	**g 79 744**	**h 148 333**

11.5 MULTIPLYING BINOMIALS

In the Vedic system we do not have one method for multiplying numbers and another for multiplying algebraic expressions. The same *Vertically and Crosswise* pattern can be used for both cases.

13 Multiply: **(x + 3)(x + 4)**.

We have to multiply x+3 by x+4.
This means that the x and the 3 in x+3 must both multiply the x and the 4 in x+4.

The best way to do this is to use the *Vertically and Crosswise* method.

Put one binomial under the other:
Multiply vertically on the left: $x \times x = \mathbf{x^2}$.
Cross-multiply and add: $4 \times x + 3 \times x = \mathbf{7x}$.
Multiply vertically on the right: $3 \times 4 = \mathbf{12}$.

$$\begin{array}{ccc} x & + & 3 \\ x & + & 4 \\ \hline x^2 & + 7x & + 12 \end{array}$$

It is just like multiplying two 2-figure numbers together.
Multiply from left to right or right to left: whichever you like.

✐ **Practice G** Multiply:

a (x + 5)(x + 6)	**b** (x + 2)(x + 9)	**c** (x + 10)(x + 1)	**d** (x + 20)(x + 20)
e (x + 1)(x + 1)	**f** (x + 22)(x + 28)	**g** (y + 52)(y + 4)	**h** $(x + 4)^2$

a $\mathbf{x^2+11x+30}$	**b** $\mathbf{x^2+11x+18}$	**c** $\mathbf{x^2+11x+10}$	**d** $\mathbf{x^2+40x+400}$
e $\mathbf{x^2+2x+1}$	**f** $\mathbf{x^2+50x+616}$	**g** $\mathbf{y^2+56y+208}$	**h** $\mathbf{x^2+8x+16}$

14 Multiply **(2x + 5)(3x + 2)**.

2x + 5	Vertically on the left: 2x×3x = $\mathbf{6x^2}$.
3x + 2	Crosswise: 4x+15x = **19x**.
$6x^2 + 19x + 10$	Vertically on the right: 5×2 = **10**.

15 Multiply **(x + 3y)(5x + 7y)**.

x + 3y	On the left: x×5x = $\mathbf{5x^2}$.
5x + 7y	Crosswise: 7xy + 15xy = **22xy**.
$5x^2 + 22xy + 21y^2$	On the right: 3y×7y = $\mathbf{21y^2}$.

✐ **Practice H** Multiply the following:

a (2x + 5)(x + 4) **b** (x + 8)(3x + 11) **c** (2x + 1)(2x + 20) **d** (2x + 3)(3x + 7)

e (4x + 3)(x + 6) **f** (3x + 17)(3x + 4) **g** (6x + 1)(5x + 1) **h** (2x + 5)(4x + 5)

i (3x + 3)(4x + 5) **j** (2x + 3y)(2x + 5y) **k** (5x + 2y)(2x + 5y) **l** (4x + 3y)(7x + y)

---

a $\mathbf{2x^2+13x+20}$ **b** $\mathbf{3x^2+35x+88}$ **c** $\mathbf{4x^2+42x+20}$ **d** $\mathbf{6x^2+23x+21}$
e $\mathbf{4x^2+27x+18}$ **f** $\mathbf{9x^2+63x+68}$ **g** $\mathbf{30x^2+11x+1}$ **h** $\mathbf{8x^2+30x+25}$
i $\mathbf{12x^2+27x+15}$ **j** $\mathbf{4x^2+16xy+15y^2}$ **k** $\mathbf{10x^2+29xy+10y^2}$ **l** $\mathbf{28x^2+25xy+3y^2}$

So, unlike the current system, we use the same method for algebraic products as for arithmetic ones.

Next we need to use the methods for combining negative numbers.

16 Multiply **(2x – 3)(3x + 4)**.

This is very similar.
2x × 3x = $\mathbf{6x^2}$.
Crosswise: 8x – 9x = –1x or **–x**.
And –3×4 = **–12**.

2x – 3
3x + 4
$6x^2 - x - 12$

17 Find **(x – 3)(x – 6)**.

Vertically: x × x = $\mathbf{x^2}$.
Crosswise: –6x –3x = **–9x**.
Vertically: –3 × –6 = **+18**.

x – 3
x – 6
$x^2 - 9x + 18$

✐ **Practice I** Multiply:

a $(x + 3)(x – 5)$ **b** $(x + 7)(x – 2)$ **c** $(x – 4)(x + 5)$ **d** $(x – 5)(x – 4)$

e $(x – 3)(x – 3)$ **f** $(2x – 3)(x + 4)$ **g** $(2x – 3)(3x + 6)$ **h** $(3x – 1)(x + 7)$

---

a $x^2–2x–15$ **b** $x^2+5x–14$ **c** $x^2+x–20$ **d** $x^2–9x+20$
e $x^2–6x+9$ **f** $2x^2+5x–12$ **g** $6x^2+3x–18$ **h** $3x^2+20x–7$

THE DIGIT SUM CHECK

The algebraic form of the digit sum check can be used.
If, for example, we wanted to check Example 14 above: $\mathbf{(2x + 5)(3x + 2) = 6x^2 + 19x + 10}$
we check that the product of the sum of the coefficients in the brackets on the left-hand side equals the sum of the coefficients on the right-hand side.

That is $(2 + 5)(3 + 2) = 6 + 19 + 10$.
Since both sides come to 35 this confirms the answer.

11.6 MULTIPLYING 3-FIGURE NUMBERS

18 Find **504 × 321**.

```
  5  0  4
  3  2  1
1 6 1 7 8 4
```

The extended pattern for multiplying 3-figure numbers is as follows.

A Vertically on the left, 5×3 = **15**.

```
  5  0  4
  |
  3  2  1
1 5
```

B Then crosswise on the left,
5×2 + 0×3 = 10.
Combining the 15 and 10 as before:
15,10 = **160**.

```
  5  0  4
     ×
  3  2  1
1 6 0
```

C Next we take 3 products and add them up,
5×1 + 0×2 + 4×3 = 17. And 160,17 = **1617**.

(actually we are gathering up the hundreds by multiplying hundreds by units, tens by tens and units by hundreds)

```
  5  0  4
  3  2  1
1 6 1 7
```

D Next we multiply crosswise on the right,
0×1 + 4×2 = 8: 1617,8 = **16178**.

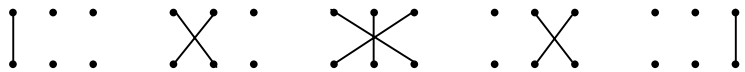
```
 5  0  4
      ×
 3  2  1
16178
```

E Finally, vertically on the right,
4×1 = 4: 16178,4 = **161784**.

```
 5  0  4
      |
 3  2  1
161784
```

Note the symmetry in the 5 steps:
first there is 1 product, then 2, then 3, then 2, then 1.

We may summarise these steps as shown below:



A **B** **C** **D** **E**

19 Find **321 × 321**.

```
 3 2 1
 3 2 1  ×
103041
```

The 5 results are 9,12,10,4,1.
The mental steps are 9
9,12 = 102
102,10 = 1030
1030,4,1 = 103041

Sometimes we have a choice about how we multiply.

20 Find **123 × 45**.

This can be done with the moving multiplier method or by the smaller vertical and crosswise pattern, treating 12 in 123 as a single digit.
Alternatively, we can put 045 for 45 and use the extended vertical and crosswise pattern:

```
1 2 3
0 4 5
5535
```

For the 5 steps we get 0,4,13,22,15.
Mentally we think 4; 53; 552; 5535.

"We thus follow a process of ascent and descent (going forward with the digits on the upper row and coming rearward with the digits on the lower row)."
From "Vedic Mathematics", Page 42.

✐ **Practice J** Multiply (there are no carries in the first few sums):

a	1 2 1		**b**	1 3 1		**c**	1 2 1		**d**	3 1 3
	1 3 1			2 1 2			2 2 2			1 2 1
e	2 1 2		**f**	1 2 3		**g**	2 1 2		**h**	2 2 2
	3 1 3			3 2 1			4 1 4			3 3 3
i	2 4 6		**j**	1 0 5		**k**	1 0 6		**l**	5 1 5
	3 3 3			5 0 7			2 2 2			5 5 5
m	4 4 4		**n**	3 2 1		**o**	1 2 3		**p**	1 2 4
	7 7 7			3 2 1			2 7 1			3 5 6

a	**15 851**	**b**	**27 772**	**c**	**26 862**	**d**	**37 873**
e	**66 356**	**f**	**39 483**	**g**	**87 768**	**h**	**73 926**
i	**81 918**	**j**	**53 235**	**k**	**23 532**	**l**	**285 825**
m	**344 988**	**n**	**103 041**	**o**	**33 333**	**p**	**44 144**

11.7 WRITTEN CALCULATIONS

It is also useful to be able to write out our multiplications.
In the Vedic system we can do this from left to right or from right to left.
Here we use the right to left method, but the formula is the same: *Vertically and Crosswise***.**

21 Find **42 × 31**.

The sum is set out as before:

A. We **multiply vertically on the right**: 2×1 = **2**,
and put this down as the right-hand figure of the answer.

B. Then we **multiply crosswise** and add to get 4+6 = **10**.
So we put down 0 and carry 1 to the left.

C. Finally we **multiply vertically on the left**: 4×3 = **12**,
12 + the carried 1 makes 13, which we put down.

```
      4 2
      3 1
  1 3 0 2
    1
```

22 Find **86 × 23**.

A. The method is as above: vertically on the right, 6×3 = **18**, put down 8 carry 1.
B. Crosswise, 24 + 12 = **36**, 36 + carried 1 = 37, put down 7 carry 3.
C. Vertically on the left, 8×2 = **16**, 16 + carried 3 = 19, put down 19.

```
        8 6
        2 3
    1 9 7 8
      3 1
```

23 Find **4321 × 24**.

Here we can use the moving multiplier method.
A. First, vertically on the right, 1×4 = 4, put it down.
B. Crosswise, 8+2 = 10, put down 0, carry 1.
C. Next we cross-multiply the 32 with the 24, this gives 12+4 = 16, 16 + carried 1 gives 17, put down 7 carry 1.
D. Then cross-multiply the 43 with the 24, this gives 16+6 = 22, 22 + carried 1 gives 23, put down 3 carry 2.
E. Vertically on the left, 4×2 = 8, 8 + carried 2 gives 10, put down 10.

```
      4 3 2 1
        2 4
  1 0 3 7 0 4
    2 1 1
```

24 Find **234 × 234**.

```
      2 3 4
      2 3 4
  5 4 7 5 6
  1 2 2 1
```

We simply do the same operations as shown in Section 11.6 but start at the right side:
4×4 = 16, put down 6 and carry 1 to the left.
3×4 + 4×3 = 24, 24 + carried 1 = 25, put down 5 and carry 2.
And so on.

Practice K Multiply the following from right to left:

a 31 × 41	**b** 23 × 22	**c** 61 × 42
d 52 × 53	**e** 54 × 45	**f** 78 × 33
g 17 × 71	**h** 88 × 88	**i** 231 × 32
j 416 × 41	**k** 182 × 23	**l** 473 × 37
m 5432 × 32	**n** 6014 × 24	**o** 3333 × 22
p 444 × 333	**q** 543 × 345	**r** 707 × 333

a	**1 271**	**b**	**506**	**c**	**2 562**
d	**2756**	**e**	**2 430**	**f**	**2 574**
g	**1 207**	**h**	**7 744**	**i**	**7 392**
j	**17 056**	**k**	**4 186**	**l**	**17 501**
m	**173 824**	**n**	**144 336**	**o**	**73 326**
p	**147852**	**q**	**187335**	**r**	**235431**

SETTING THE SUMS OUT

In Example 24 each of the five steps had a center of symmetry.
The five dots on the right show these five centers and as we move from left to right or right to left through the sum it is as if there is a dot moving through the sum.

```
 2  3  4
 •  •  •  •  •
 2  3  4  ×
54756
```

In the calculation shown here the units figure of the result of each of the five steps is placed under the dot for that step.

Other ways of setting the sums and answers out are possible and may be preferred.

"On seeing this kind of work actually being performed by the little children, the doctors, professors and other "big-guns" of mathematics are wonder struck and exclaim: "Is this mathematics or magic?" And we invariably answer and say: "It is both. It is magic until you understand it; and it is mathematics thereafter"; and then we proceed to substantiate and prove the correctness of this reply of ours!"
From "Vedic Mathematics", Page xvii.